

BIOINFORMATICS

WEEK 2 – DAY 9

GENOME

The genome is the entire archive of genetic material inside an organism.

- Think of it as a complete master library.
- It includes all functional genes and non-coding sequence regions.
- It is identical in almost every single cell of your body.

CHROMOSOMES

A chromosome is a single, long strand of DNA tightly wrapped around proteins.

- Think of them as individual volumes or books in that library.
- Humans have 23 pairs of chromosomes (46 total) in each cell.
- They keep the massive genome physically organized and protected during cell division.

GENE LOCUS

A locus (plural: *loci*) is the exact physical address of a gene on a specific chromosome.

- Think of it as the shelf location or page number for a specific recipe.
- For example, the human insulin (INS) gene locus is specifically at 11p15.5.
- This address means: Chromosome 11, short arm ("p"), region 1, band 5, sub-band 5.

EXONS AND INTRONS

When you open a gene, the genetic text is split into two types of segments:

- **Exons:** The expressed, coding regions of the gene. They contain the actual recipes for building proteins.
- **Introns:** The intervening, non-coding regions between exons. They act like commercial breaks or filler text that must be removed before the recipe is used.

TRANSCRIPTS

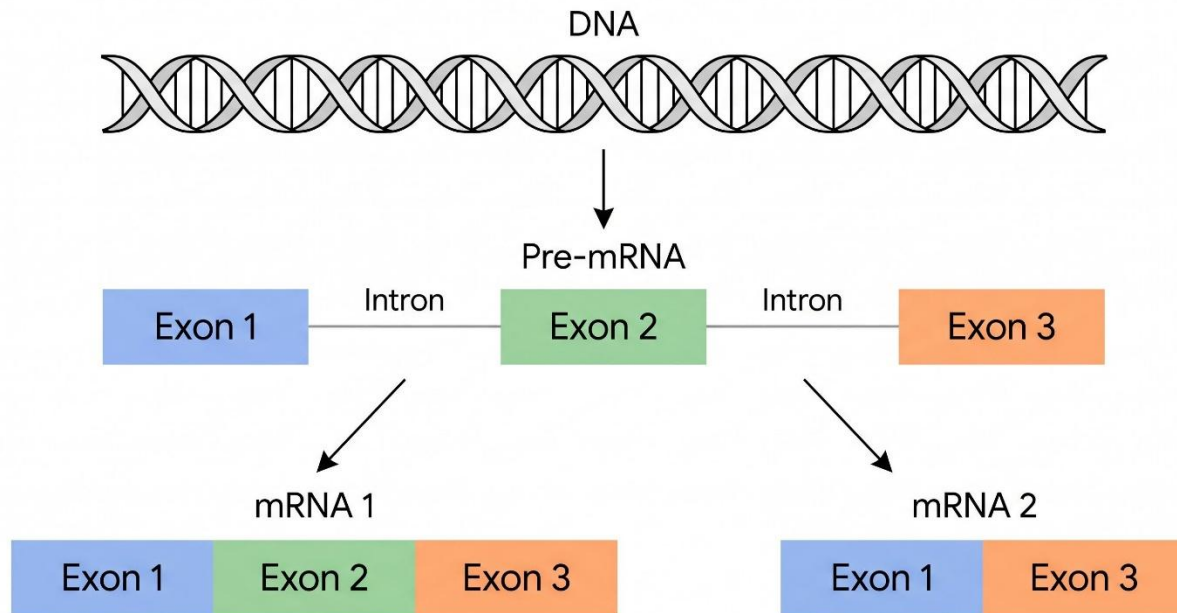
A transcript is the temporary RNA copy made from reading a DNA gene during transcription.

- **Pre-mRNA Transcript:** A raw, copy-pasted strand containing *both* exons and introns.
- **Mature mRNA Transcript:** The finalized copy after a cellular process called splicing cuts out all the introns and glues the exons back together. This clean transcript leaves the nucleus to build proteins

ISOFORMS

Isoforms are different structural versions of a protein generated from the exact same gene.

- This happens through alternative splicing, where the cell selectively mixes and matches different exons.
- For example, a gene with exons 1-2-3-4 can be spliced into transcript version A (1-2-4) or transcript version B (1-2-3).
- This enables a single gene address to produce multiple distinct molecular tools depending on tissue needs.



Ensembl

It is a premier, open-access genome browser and bioinformatics database registry created through a joint scientific partnership between the European Bioinformatics Institute (EMBL-EBI) and the Wellcome Sanger Institute.

It is designed to automate the annotation of vertebrate genomes, map phenotypic data, and provide integrated tools for comparative genomic studies.

Core Features of Ensembl

- **Automated Gene Annotation:** It combines computational prediction pipelines with curated public records to systematically locate and map gene loci, exons, introns, and regulatory sequences across hundreds of species.
- **Ensembl Stable Identifiers:** To ensure global reproducibility, Ensembl uses a strict, standard nomenclature for its records:
 - ENSG... → **Ensembl Gene ID** (e.g., ENSG00000254647 for human insulin).
 - ENST... → **Ensembl Transcript ID** (representing specific splice variants/isoforms).
 - ENSP... → **Ensembl Protein ID** (representing the resulting amino acid product).

- **Ensembl Genomes:** A dedicated sub-registry that extends this automated annotation system beyond vertebrates to include plants, fungi, bacteria, protists, and invertebrate vectors.

With a genome browser, you can determine:

- Which gene is affected.
- Whether the mutation falls in an exon or intron.
- Which transcript it affects.
- What nearby genomic features exist.